KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 8 MATHEMATICS · YEARLY EXAM

Exam preparation booklet

Key ideas · worked examples · exam-style practice · answers

Equations
20 marks · 25 min

Investigating Data
23 marks · 25 min

Mixed Topics
22 marks · 25 min

Name: _____

Teacher: _____

How to use this booklet

This booklet follows the **Year 8 Mathematics Yearly Examination** section by section. Every skill on the exam skills checklist has **key ideas**, a **worked example** and **practice questions** in the same style as the exam. Answers are at the back – try each question first, then check.

Section	Marks	Suggested time
A · Equations	20	25 minutes
B · Investigating Data	23	25 minutes
C · Mixed Topics	22	25 minutes
Total	**65**	**75 minutes**

Exam technique

- **About one minute per mark.** If you are stuck, move on and come back.
- **Show your working.** Marks are given for method, not just the final answer.
- **Calculators are permitted** – but estimate first so you notice a wrong button press.
- **Read graphs carefully:** check the scale, the key and the units before you answer.
- **Underline key words** such as *at least*, *more than*, *altogether* and *how many more*.
- **Bring:** black pen, pencil, ruler, eraser and a calculator.
- **Check:** substitute your answer back into equations; make sure answers have units.

Key words in exam questions

These instruction words tell you exactly what the marker is looking for.

Word	What you need to do	Example in the exam
Solve	Find the value of the pronumeral that makes the equation true	Q1–Q3, Q5(b), Q6(b)
Show / use substitution to show	Write every step so the marker can follow it, then write a conclusion	Q4, Q15(b)
Write an equation	Use a pronumeral and an equals sign – do not solve it yet	Q5(a), Q6(a)
Calculate / find	Work out a value. Show working when there are 2 or more marks	Q9, Q13, Q14, Q20
Identify	Point out the value or item – no working needed	Q14(a)
Evaluate	Work out the numerical value	Q17, Q19(c)
Simplify	Write in the simplest form (collect like terms, smallest whole numbers)	Q18(b), Q19(a), Q22(a)
Expand	Remove the brackets by multiplying every term inside	Q19(b)
Complete the table	Fill in every empty cell	Q21

Self-check: how confident am I?

Tick one box for each skill before you start. Spend most of your revision time on the skills you tick "Help!". The **Booklet** column tells you where to find each skill.

Skill	Booklet	Exam Q	Got it	Nearly	Help!
Section A – Equations					
Solve one-step equations using inverse operations (+, −, ×, ÷)	A1	Q1	☐	☐	☐
Solve two-step equations, including ones with division	A2	Q2	☐	☐	☐
Solve equations that give a negative solution	A3	Q3	☐	☐	☐
Check a solution by substituting it back into the equation	A4	Q4	☐	☐	☐
Turn a worded "think of a number" problem into an equation	A5	Q5a	☐	☐	☐
Write an equation for a real-life situation using a pronumeral	A6	Q6a	☐	☐	☐
Solve equations made from worded problems and interpret the answer	A5, A6	Q5b, Q6b	☐	☐	☐
Section B – Investigating Data					
Read and interpret column graphs, including finding totals	B1	Q7	☐	☐	☐
Read values from a line graph and work out changes between points	B2	Q8	☐	☐	☐
Calculate amounts from percentages in a sector (pie) graph	B3	Q9a	☐	☐	☐
Compare categories in a sector graph	B3	Q9b	☐	☐	☐
Read a stem-and-leaf plot using its key	B4	Q10a, c	☐	☐	☐
Find the median from a stem-and-leaf plot	B4	Q10b	☐	☐	☐
Read a dot plot and find the mode	B5	Q11	☐	☐	☐
Interpret "at least" and "more than" in data questions	B6	Q10c, Q11b, Q12c	☐	☐	☐
Read a frequency table, find totals and the mode	B7	Q12	☐	☐	☐
Calculate the mean of a data set	B8	Q13a, Q14b	☐	☐	☐
Find the median of a data set with an even number of values	B9	Q13b	☐	☐	☐
Find the mode and range of a data set	B10	Q13c, d	☐	☐	☐
Identify an outlier and understand its effect on the mean	B11	Q14	☐	☐	☐
Section C – Mixed Topics					
Use Pythagoras' theorem to find the hypotenuse	C1	Q15a	☐	☐	☐
Use Pythagoras' theorem to test whether a triangle is right-angled	C2	Q15b	☐	☐	☐
Multiply and divide decimals	C3	Q16	☐	☐	☐
Evaluate powers and square roots	C4	Q17	☐	☐	☐
Identify surds	C5	Q18a	☐	☐	☐
Simplify surds using square factors, e.g. √12 = 2√3	C6	Q18b	☐	☐	☐
Simplify expressions by collecting like terms	C7	Q19a	☐	☐	☐
Expand brackets using the distributive law	C8	Q19b	☐	☐	☐
Substitute values into algebraic expressions, including squared terms	C9	Q19c	☐	☐	☐
Find the area of a trapezium using the formula	C10	Q20a	☐	☐	☐
Find the volume of a rectangular prism	C11	Q20b	☐	☐	☐
Complete a table of values for a linear rule, including negative x-values	C12	Q21	☐	☐	☐
Simplify ratios	C13	Q22a	☐	☐	☐
Calculate average speed as a rate	C14	Q22b	☐	☐	☐

SECTION A

Equations

25 minutes | 20 marks

Solving one- and two-step equations, checking solutions and writing equations from worded problems. Questions 1–6 in the exam.

Skills in this section

- One-step equations using inverse operations
- Two-step equations, including division
- Equations with a negative solution
- Checking a solution by substitution
- "Think of a number" problems
- Writing and solving equations for real-life situations

A1 One-step equations and inverse operations

KEY IDEAS

Undo what has been done to the pronumeral

Operation	+	−	×	÷
Inverse (undo)	−	+	÷	×

An equation is a **balance**: whatever you do to one side, do to the other. Use the **inverse** (opposite) operation to get the pronumeral on its own.

WORKED EXAMPLE

(a) $x + 8 = 13 \rightarrow x = 13 - 8 \rightarrow \mathbf{x = 5}$ (b) $m - 6 = 11 \rightarrow m = 11 + 6 \rightarrow \mathbf{m = 17}$

(c) $7k = 56 \rightarrow k = 56 \div 7 \rightarrow \mathbf{k = 8}$ (d) $y \div 4 = 6 \rightarrow y = 6 \times 4 \rightarrow \mathbf{y = 24}$

IN THE EXAM Section A, Q1 (4 marks)

Practice – solve

(a) $a + 5 = 12$ **(b)** $x + 14 = 30$ **(c)** $n - 7 = 8$ **(d)** $p - 12 = 9$

(e) $5x = 45$ **(f)** $8m = 64$ **(g)** $k \div 3 = 9$ **(h)** $w \div 6 = 5$

A2 Two-step equations

KEY IDEAS

Undo in reverse order

Think about how the equation was built, then undo it **backwards**. Usually undo **+ or −** first, then **× or ÷**.

For $x \div 3 + 2 = 9$: the 2 was added last, so subtract 2 first, *then* multiply by 3.

WORKED EXAMPLE

Multiplication: $2x + 7 = 19$ $\underline{-7}$ $2x = 12$ $\underline{\div 2}$ $\mathbf{x = 6}$

Division: $x \div 4 - 3 = 2$ $\underline{+3}$ $x \div 4 = 5$ $\underline{\times 4}$ $\mathbf{x = 20}$

Check: $20 \div 4 - 3 = 5 - 3 = 2$ ✓

IN THE EXAM Section A, Q2 (6 marks)

Practice – solve, showing every step

(a) $4x + 3 = 23$ **(b)** $5m + 2 = 37$ **(c)** $3y - 4 = 14$ **(d)** $6k - 5 = 43$

(e) $x \div 2 + 5 = 11$ **(f)** $a \div 5 - 1 = 3$ **(g)** $n \div 3 + 7 = 10$ **(h)** $7p - 9 = 26$

A3 Equations with a negative solution

KEY IDEAS

Same steps – watch the signs

If you subtract a bigger number from a smaller one, the answer is **negative**: $4 - 10 = -6$.

A negative divided by a positive is negative: $-6 \div 3 = -2$. Use the **(–)** key on your calculator.

WORKED EXAMPLE

Solve $3x + 10 = 4$

$3x = 4 - 10$ (subtract 10)

$3x = -6$

$x = -6 \div 3$ (divide by 3)

$\boldsymbol{x = -2}$

Check: $3 \times (-2) + 10 = -6 + 10 = 4$ ✓

IN THE EXAM Section A, Q3 (2 marks)

Practice – solve

(a) $x + 9 = 4$ **(b)** $m + 12 = 5$ **(c)** $4a = -20$ **(d)** $2y + 11 = 3$

(e) $5k + 8 = -7$ **(f)** $3n - 2 = -14$ **(g)** $x \div 2 + 6 = 1$ **(h)** $6p + 20 = 2$

A4 Checking a solution by substitution

KEY IDEAS

Substitute, work out, compare

1. Replace the pronumeral with the value (use brackets for negatives).
2. Work out the **left-hand side (LHS)**.
3. Compare it with the **right-hand side (RHS)**. If LHS = RHS, the value **is** the solution.

Always finish with a sentence that answers the question.

WORKED EXAMPLE

(a) Is $x = 5$ the solution to $4x - 3 = 17$?

LHS = $4 \times 5 - 3 = 20 - 3 = 17$ = RHS ✓ → **Yes**, $x = 5$ is the solution.

(b) Is $m = 3$ the solution to $2m + 7 = 15$?

LHS = $2 \times 3 + 7 = 6 + 7 = 13 \neq 15$ → **No**, $m = 3$ is not the solution.

IN THE EXAM Section A, Q4 (2 marks)

Practice – use substitution to show whether each value is the solution

(a) $x = 6$ in $2x + 5 = 17$ **(b)** $y = 4$ in $5y - 8 = 14$ **(c)** $k = 9$ in $k \div 3 + 4 = 7$

(d) $m = -2$ in $3m + 10 = 4$ **(e)** $n = 7$ in $6n - 12 = 32$ **(f)** $a = 8$ in $a \div 2 - 3 = 1$

A5 "Think of a number" problems

KEY IDEAS

Turn the words into symbols, in order

Words	Symbols
I think of a number	n
multiply it by 4	$4n$
then add 5	$4n + 5$
divide it by 2	$n \div 2$
subtract 6	$- 6$
the answer is … / is equal to …	$=$

Write the equation **first**, then solve it as a two-step equation. Check your number in the original words.

WORKED EXAMPLE

I think of a number, multiply it by 3 and then add 8. The answer is 29.

Equation: $\mathbf{3n + 8 = 29}$

$3n = 21$ (subtract 8) → $\boldsymbol{n = 7}$ (divide by 3)

Check: $7 \times 3 = 21$, $21 + 8 = 29$ ✓

IN THE EXAM Section A, Q5 (3 marks)

Practice – write an equation using *n*, then solve it

(a) I think of a number and add 15. The answer is 32.

(b) I think of a number, multiply it by 5, and then add 4. The answer is 39.

(c) I think of a number, multiply it by 6, and then subtract 10. The answer is 44.

(d) I think of a number, divide it by 2, and then add 9. The answer is 15.

(e) I think of a number, multiply it by 3, and then subtract 5. The answer is −11.

A6 Writing and solving equations for real-life situations

KEY IDEAS

Unknown → pronumeral, then build the equation

1. **Define** the pronumeral, with units (e.g. let p be the cost of one pen in dollars).
2. Things that depend on the unknown are **multiplied** (4 drinks → $4d$). Fixed amounts are **added**.
3. The **total** goes on the other side of the equals sign.
4. Solve, then **interpret**: answer in a sentence with units, and check it makes sense.

WORKED EXAMPLE

Jack buys 4 drinks that each cost the same and a sandwich that costs $7. He pays $19 altogether. Find the cost of one drink.

Let d = cost of one drink ($). Equation: $\mathbf{4d + 7 = 19}$

$4d = 12 \rightarrow d = 3 \rightarrow$ **One drink costs $3.**

Check: 4 × $3 + $7 = $19 ✓

IN THE EXAM Section A, Q6 (3 marks) – also Q5(b)

Practice – write an equation, solve it and answer in a sentence

(a) Ella buys 3 notebooks that each cost the same and a pencil case that costs $8. She spends $23. Find the cost of one notebook.

(b) A taxi charges $4 to start plus $2 for every kilometre. A trip costs $26. How many kilometres long was the trip?

(c) A gym charges a $15 joining fee plus $10 per visit. Sam has paid $85 so far. How many times has he visited?

(d) Six friends share the cost of a pizza order equally, and each also pays $2 for a drink. Each person pays $9 in total. What did the pizza order cost?

(e) A rectangle has a length of 9 cm and a perimeter of 30 cm. Write an equation using w for the width and find the width.

SECTION B

Investigating Data

25 minutes | 23 marks

Reading graphs and displays, and calculating summary statistics. Questions 7–14 in the exam.

Skills in this section

- Column, line and sector (pie) graphs
- Stem-and-leaf plots and dot plots
- Frequency tables; “at least” and “more than”
- Mean, median (odd and even), mode and range
- Outliers and their effect on the mean

Reading graphs and displays

B1 Column graphs

KEY IDEAS

Read across from the top of each column

Check the scale first – a gridline might be worth 1, 2, 5 or more. A column that stops halfway between two gridlines is halfway between their values.

Most popular = tallest column. **Total surveyed** = add the heights of **all** the columns.

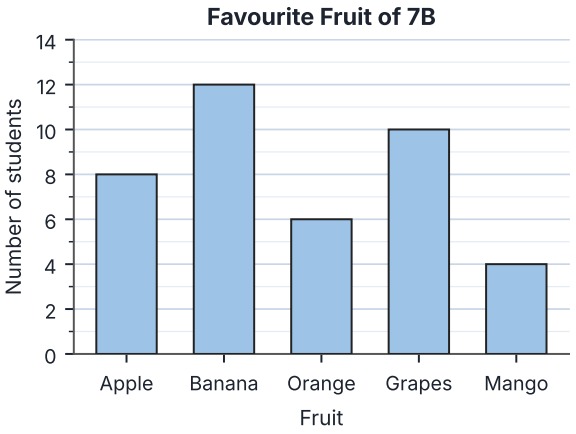
WORKED EXAMPLE

The graph shows the favourite fruit of a Year 7 class.

Most popular: tallest column → **Banana** (12 students)

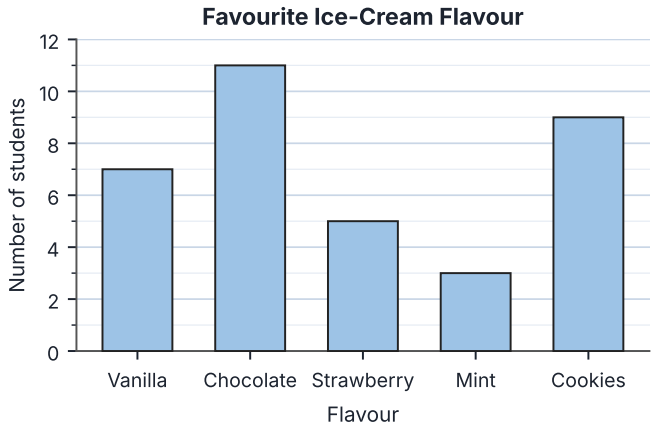
Total surveyed: 8 + 12 + 6 + 10 + 4 = **40 students**

How many more chose banana than orange? 12 − 6 = **6**



IN THE EXAM Section B, Q7 (2 marks)

Practice



(a) Which flavour was the most popular?

(b) Which flavour was the least popular?

(c) How many students were surveyed altogether?

(d) How many more students chose chocolate than strawberry?

(e) What fraction of the students chose vanilla? Simplify your answer.

B2 Line graphs

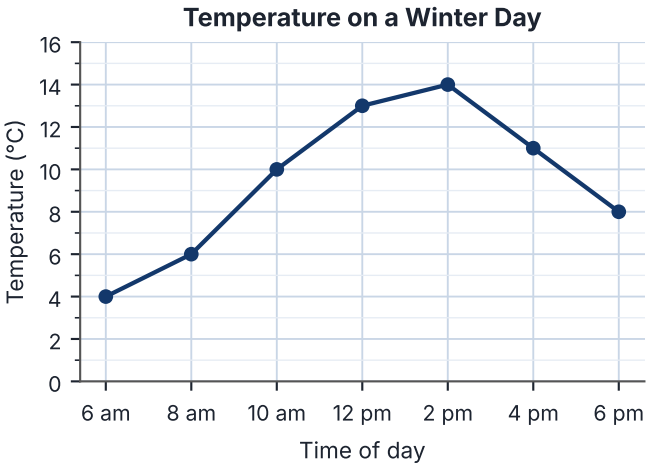
KEY IDEAS

Find the time, go up to the line, then across

Find the time on the horizontal axis, go **up** to the line, then **across** to read the value on the vertical axis.

Highest value = highest point. **Change** between two times = later value − earlier value (a rise if positive, a fall if negative).

WORKED EXAMPLE



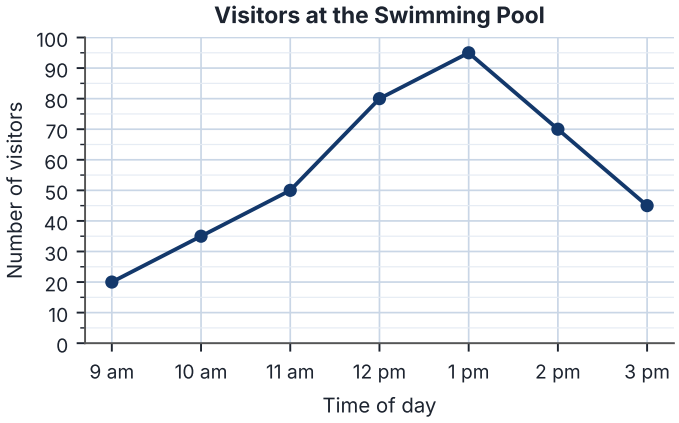
Highest temperature: highest point → **2 pm** (14 °C)

Rise from 6 am to 12 pm:
13 − 4 = **9 °C**

Fall from 2 pm to 6 pm:
14 − 8 = **6 °C**

IN THE EXAM Section B, Q8 (2 marks)

Practice



(a) At what time was the pool busiest?

(b) How many visitors were there at 11 am?

(c) By how much did the number of visitors increase between 9 am and 12 pm?

(d) By how much did the number of visitors decrease between 1 pm and 3 pm?

(e) Between which two times shown did the number of visitors rise the most?

B3 Sector (pie) graphs

KEY IDEAS

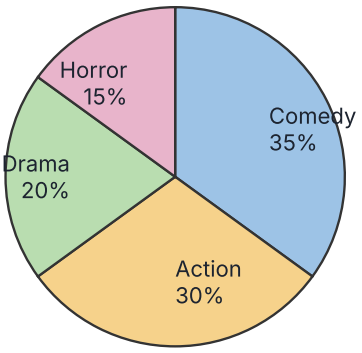
The whole circle is 100% of the total

Amount = percentage × total. For example, 40% of 120 = 0.4 × 120 = 48 (or 40 ÷ 100 × 120).

Comparing two sectors: either find both amounts and subtract, or subtract the percentages first and find that percentage of the total. The biggest sector is the most common category.

WORKED EXAMPLE

Favourite Movie Type of 200 People



200 people were asked their favourite type of movie.

How many chose comedy?

35% × 200 = 0.35 × 200 = **70 people**

How many more chose action than horror?

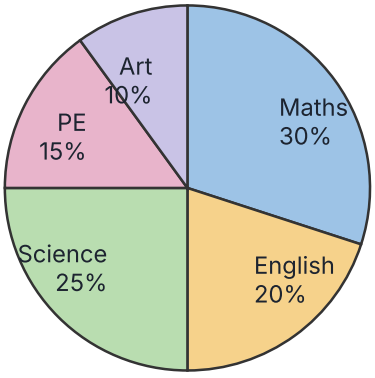
30% − 15% = 15% → 0.15 × 200 = **30 more people**

Check: action 60, horror 30, 60 − 30 = 30 ✓

IN THE EXAM Section B, Q9 (3 marks)

Practice

Favourite Subject of 80 Students



(a) Which subject was the most popular?

(b) How many students chose Science?

(c) How many students chose PE?

(d) How many more students chose Maths than Art?

(e) How many students chose Science or PE?

(f) How many students did **not** choose English?

B4 Stem-and-leaf plots and the median

KEY IDEAS

Stem = tens, leaf = units – always read the key

Each **leaf** is one data value, so **count the leaves** to find how many values there are. The values are already in order from smallest to largest.

Median = the middle value. For n values it is in position $\frac{n+1}{2}$. With 15 values, the median is the 8th value.

WORKED EXAMPLE

The ages of 11 people at a party.

Number of people: 3 + 4 + 3 + 1 = **11**

Median position: $\frac{11+1}{2}$ = 6th value

12, 15, 18, 20, 23, **23** … → median = **23**

Oldest: 42 · Youngest: 12

Stem	Leaf
1	2 5 8
2	0 3 3 7
3	1 4 6
4	2

Key: 1 | 2 means 12

IN THE EXAM Section B, Q10 (3 marks)

Practice – points scored by a basketball team

Stem	Leaf
1	4 8
2	0 2 5 5 9
3	1 3 6 8
4	0 5

Key: 1 | 4 means 14 points

(a) How many games were played?

(b) What was the highest score?

(c) Find the median score.

(d) Find the range.

(e) In how many games did they score more than 30 points?

(f) In how many games did they score at least 25 points?

B5 Dot plots and the mode

KEY IDEAS

Each dot is one data value

The number of dots above a value is how many times that value occurred. The total number of dots = the number of data values.

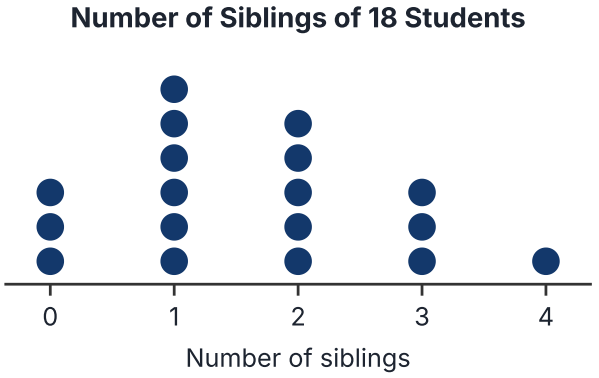
The **mode** is the value with the **tallest stack** of dots. Give the value from the axis, not the number of dots.

WORKED EXAMPLE

Mode: tallest stack is above 1 → **mode = 1 sibling**

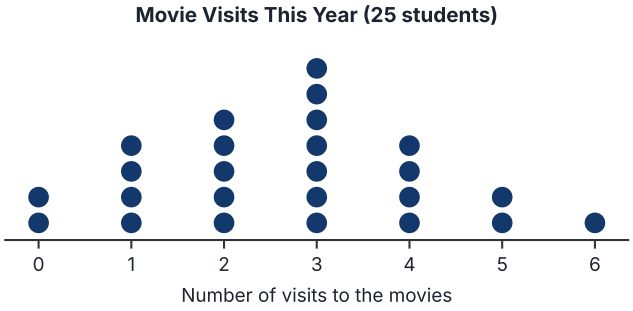
Check the total: 3 + 6 + 5 + 3 + 1 = 18 ✓

How many have 3 siblings? **3 students**



IN THE EXAM Section B, Q11 (2 marks)

Practice



(a) How many students went to the movies exactly twice?

(b) What is the mode?

(c) How many students went at least 4 times?

(d) How many students went more than 4 times?

(e) How many students went fewer than 2 times?

B6 "At least" and "more than"

KEY IDEAS

Decide whether the boundary value is included

Phrase	Means	Includes 4?
more than 4	5, 6, 7, …	**No**
at least 4	4, 5, 6, …	**Yes**
fewer than / less than 4	… 2, 3	**No**
at most 4	… 2, 3, 4	**Yes**

Underline the phrase in the question, decide which values count, then add them up.

WORKED EXAMPLE

Using the siblings dot plot in B5:

At least 2 siblings → 2, 3 or 4: 5 + 3 + 1 = **9 students**

More than 2 siblings → 3 or 4: 3 + 1 = **4 students**

IN THE EXAM Section B, Q10(c), Q11(b) and Q12(c)

Practice – quiz scores out of 10 for 12 students: 4, 7, 8, 6, 10, 7, 5, 9, 7, 3, 8, 6

(a) How many students scored more than 7?

(b) How many students scored at least 7?

(c) How many students scored fewer than 6?

(d) How many students scored at most 6?

(e) How many students scored at least 9?

(f) How many students scored exactly 7?

B7 Frequency tables

KEY IDEAS

Frequency = how many times

Total = add the **frequency** row (not the values row).

Mode = the value with the **highest frequency**. Give the value, not the frequency.

For "at least" or "more than", add the frequencies of the values that count.

WORKED EXAMPLE

The number of people in each car passing a school.

People in car	1	2	3	4	5
Frequency	12	8	5	3	2

Total cars: 12 + 8 + 5 + 3 + 2 = **30** Mode: highest frequency is 12 → **1 person**

At least 3 people: 5 + 3 + 2 = **10 cars**

IN THE EXAM Section B, Q12 (3 marks)

Practice – goals scored by a soccer team in each game

Goals	0	1	2	3	4	5
Frequency	3	6	8	4	2	1

(a) How many games were played?

(b) What is the mode?

(c) In how many games did they score no goals?

(d) In how many games did they score at least 3 goals?

(e) In how many games did they score more than 1 goal?

(f) In how many games did they score fewer than 2 goals?

Summary statistics

B8 The mean

KEY IDEAS

Add them up, divide by how many

$$\text{Mean} = \frac{\text{sum of the values}}{\text{number of values}}$$

The mean does not have to be one of the data values, and can be a decimal. Check it is between the smallest and largest values.

WORKED EXAMPLE

Find the mean of 6, 9, 4, 8, 7, 8.

Sum = 6 + 9 + 4 + 8 + 7 + 8 = 42 Number of values = 6

Mean = $\frac{42}{6}$ = **7**

IN THE EXAM Section B, Q13(a) and Q14(b)

Practice – find the mean (round to 1 decimal place if needed)

(a) 5, 8, 6, 9, 7

(b) 12, 15, 10, 18, 20

(c) 3, 7, 2, 8, 5, 5

(d) 21, 25, 19, 24, 26, 23

(e) 4.5, 6, 5.5, 7, 8

(f) 13, 17, 11, 15, 19, 12, 18, 14

B9 The median of an even number of values

KEY IDEAS

Order first, then find the middle

1. Write the values in **order** from smallest to largest.
2. **Odd** number of values → the median is the middle value.
3. **Even** number of values → there are two middle values. Add them and divide by 2.

WORKED EXAMPLE

Find the median of 9, 4, 12, 7, 15, 10.

In order: 4, 7, **9, 10**, 12, 15 (6 values, so two in the middle)

Median = $\frac{9 + 10}{2}$ = **9.5**

IN THE EXAM Section B, Q13(b) (1 mark)

Practice – find the median

(a) 3, 8, 5, 6

(b) 12, 7, 10, 15, 9, 11

(c) 20, 14, 18, 16, 22, 18

(d) 6, 2, 9, 4, 7, 1, 8, 5

(e) 31, 25, 28, 40, 22, 35, 27, 30

(f) 13, 9, 17, 11, 15

B10 The mode and range

KEY IDEAS

Most common, and the spread

Mode = the value that occurs most often. There can be more than one mode, or no mode.
Range = highest value − lowest value. The range is a single number (not "3 to 12").

WORKED EXAMPLE

For 7, 3, 9, 7, 5, 12, 7, 4:

Mode: 7 appears three times → **mode = 7** Range: 12 − 3 = **9**

IN THE EXAM Section B, Q13(c) and Q13(d) (2 marks)

Practice – find the mode and the range

(a) 4, 6, 4, 8, 9, 4, 5

(b) 15, 22, 18, 22, 30, 11

(c) 2.5, 3.1, 2.8, 3.1, 2.2

(d) 10, 12, 10, 15, 12, 18

(e) 45, 51, 38, 47, 60

B11 Outliers and their effect on the mean

KEY IDEAS

An outlier is much bigger or much smaller than the rest

A **large** outlier pulls the mean **up**. A **small** outlier pulls the mean **down**.

The median is hardly affected, so when there is an outlier the median often describes a "typical" value better than the mean.

WORKED EXAMPLE

Hours of sleep: 8, 7, 9, 8, 2, 8

Outlier: **2** (much lower than the others)

Mean with the outlier: $\frac{42}{6}$ = **7 hours** Mean without: $\frac{40}{5}$ = **8 hours**

The outlier pulled the mean **down** by 1 hour.

IN THE EXAM Section B, Q14 (3 marks)

Practice

1. The prices of six sandwiches at a café are: $8, $9, $7, $10, $8, $30

(a) Identify the outlier.

(b) Calculate the mean price, including the outlier.

(c) Calculate the mean price without the outlier.

(d) Explain the effect of the outlier on the mean.

2. A student's spelling test scores are: 18, 17, 19, 3, 18, 20, 17

(a) Identify the outlier.

(b) Calculate the mean, including the outlier.

(c) Calculate the mean without the outlier (1 decimal place).

(d) Find the median with and without the outlier. What do you notice?

SECTION C

Mixed Topics

25 minutes | 22 marks

Pythagoras' theorem, number skills, algebra, measurement, linear relationships, ratios and rates. Questions 15–22 in the exam.

Skills in this section

- Pythagoras' theorem: hypotenuse and testing for a right angle
- Decimals, powers, square roots and surds
- Algebra: like terms, expanding, substitution
- Area of a trapezium and volume of a rectangular prism
- Tables of values for linear rules
- Simplifying ratios and average speed

Pythagoras' theorem

C1 Finding the hypotenuse

KEY IDEAS

$c^2 = a^2 + b^2$

The **hypotenuse** c is the longest side, opposite the right angle. a and b are the two shorter sides.

Square the two shorter sides, **add**, then take the **square root**. Round to 1 decimal place if the answer is not exact.

WORKED EXAMPLE

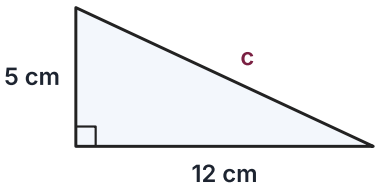
$c^2 = 5^2 + 12^2$

$c^2 = 25 + 144$

$c^2 = 169$

$c = \sqrt{169}$ = **13 cm**

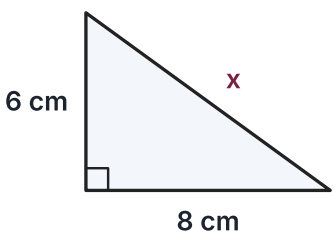
Not exact? Sides 6 cm and 9 cm: $c^2 = 36 + 81 = 117$, $c = \sqrt{117}$ = 10.816... ≈ **10.8 cm**



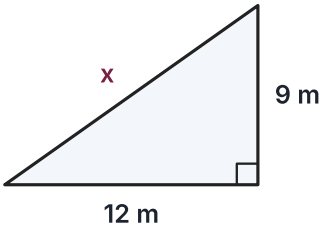
IN THE EXAM Section C, Q15(a) (2 marks)

Practice – find the length of the hypotenuse (1 decimal place where needed)

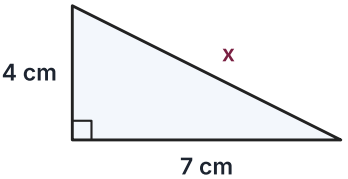
(a)



(b)



(c)



(d) The two shorter sides of a right-angled triangle are 10 cm and 15 cm. Find the hypotenuse.

(e) A TV screen is 80 cm wide and 45 cm high. How long is its diagonal?

C2 Testing whether a triangle is right-angled

KEY IDEAS

Does longest² = the sum of the other two squares?

1. Square the **longest** side.
2. Square the two shorter sides and **add**.
3. If the answers are **equal**, the triangle is right-angled (the right angle is opposite the longest side). If not, it is not right-angled.

Useful Pythagorean triads: 3, 4, 5 · 5, 12, 13 · and their multiples (6, 8, 10 · 9, 12, 15).

WORKED EXAMPLE

(a) 9 cm, 12 cm, 15 cm: $15^2 = 225$ and $9^2 + 12^2 = 81 + 144 = 225$
Equal, so **yes, it is right-angled**.

(b) 6 cm, 7 cm, 9 cm: $9^2 = 81$ and $6^2 + 7^2 = 36 + 49 = 85$
$81 \neq 85$, so **no, it is not right-angled**.

IN THE EXAM Section C, Q15(b) (2 marks)

Practice – is each triangle right-angled? Show working

(a) 5 cm, 12 cm, 13 cm
(b) 6 cm, 8 cm, 11 cm
(c) 12 m, 35 m, 37 m
(d) 10 cm, 24 cm, 26 cm
(e) 4 m, 6 m, 7 m
(f) 9 cm, 40 cm, 41 cm

Number

C3 Multiplying and dividing decimals

KEY IDEAS

Count decimal places; make the divisor a whole number

Multiplying: ignore the decimal points and multiply, then count the total number of decimal places in the question and put that many in the answer.

Dividing by a decimal: multiply **both** numbers by 10, 100, … until you are dividing by a whole number. Estimate to check your answer is sensible.

WORKED EXAMPLE

(a) 3.6 × 0.4: 36 × 4 = 144; two decimal places → **1.44** (estimate 4 × 0.4 = 1.6 ✓)

(b) 5.4 ÷ 0.09: multiply both by 100 → 540 ÷ 9 = **60**

IN THE EXAM Section C, Q16 (2 marks)

Practice – calculate

(a) 2.5 × 0.6
(b) 1.2 × 0.3
(c) 6.4 × 0.25
(d) 0.7 × 0.08
(e) 4.8 ÷ 0.6
(f) 3.6 ÷ 0.04
(g) 8.4 ÷ 0.07
(h) 12.5 ÷ 0.5

C4 Powers and square roots

KEY IDEAS

A power is repeated multiplication

$2^4 = 2 \times 2 \times 2 \times 2 = 16$ (**not** 2×4). The small number tells you how many times the base is multiplied.

$\sqrt{64}$ asks "what number squared is 64?" → 8, because $8^2 = 64$.

Perfect squares to know: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144.

WORKED EXAMPLE

(a) $3^4 = 3 \times 3 \times 3 \times 3 =$ **81** (b) $\sqrt{81} =$ **9** because $9^2 = 81$ (c) $5^2 + \sqrt{49} = 25 + 7 =$ **32**

IN THE EXAM Section C, Q17 (2 marks)

Practice – evaluate

(a) 2^5 **(b)** 3^3 **(c)** 10^4 **(d)** 5^3 **(e)** $\sqrt{36}$

(f) $\sqrt{121}$ **(g)** $\sqrt{144}$ **(h)** $4^2 - \sqrt{25}$ **(i)** $(-3)^2$ **(j)** $\sqrt{100} \times 2^3$

C5 Identifying surds

KEY IDEAS

A surd is a square root that is not a whole number

If the number under the square root is a **perfect square**, it is **not** a surd: $\sqrt{16} = 4$.

If it is not a perfect square, the answer is a decimal that never ends or repeats, so it **is** a surd: $\sqrt{7} = 2.6457...$

WORKED EXAMPLE

$\sqrt{36} = 6$ → **not a surd** $\sqrt{20} = 4.472...$ → 20 is not a perfect square → **surd**

IN THE EXAM Section C, Q18(a) (1 mark)

Practice – circle the surds in each row

(a) $\sqrt{4}$ $\sqrt{5}$ $\sqrt{49}$ $\sqrt{30}$

(b) $\sqrt{81}$ $\sqrt{18}$ $\sqrt{100}$ $\sqrt{2}$

(c) $\sqrt{1}$ $\sqrt{45}$ $\sqrt{64}$ $\sqrt{15}$

(d) $\sqrt{72}$ $\sqrt{144}$ $\sqrt{3}$ $\sqrt{169}$

C6 Simplifying surds

KEY IDEAS

Find the largest square factor

Use $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$. Write the number as **(largest square factor) × (what is left)**, then take the square root of the square.

Square factors to look for: 4, 9, 16, 25, 36, 49, 64, 81, 100. Check on your calculator that both forms give the same decimal.

WORKED EXAMPLE

(a) $\sqrt{12} = \sqrt{4 \times 3} = \sqrt{4} \times \sqrt{3} =$ **$2\sqrt{3}$**

(b) $\sqrt{72} = \sqrt{36 \times 2} =$ **$6\sqrt{2}$** (using 9×8 gives $3\sqrt{8}$, which can still be simplified – always use the largest square factor)

IN THE EXAM Section C, Q18(b) (2 marks)

Practice – simplify

(a) $\sqrt{8}$ **(b)** $\sqrt{18}$ **(c)** $\sqrt{20}$ **(d)** $\sqrt{27}$ **(e)** $\sqrt{45}$

(f) $\sqrt{48}$ **(g)** $\sqrt{75}$ **(h)** $\sqrt{98}$ **(i)** $\sqrt{200}$

Algebra

C7 Collecting like terms

KEY IDEAS

Only like terms can be added or subtracted

Like terms have exactly the same pronumerals: $3x$ and $-5x$ are like terms; $3x$ and $3y$ are not. Add or subtract the numbers in front (the coefficients) and keep the pronumeral. The sign in front of a term belongs to it.

WORKED EXAMPLE

Simplify $6x + 2y - 4x + 3y$

$= 6x - 4x + 2y + 3y$ (group like terms)

$=$ **$2x + 5y$**

IN THE EXAM Section C, Q19(a) (1 mark)

Practice – simplify

(a) $5a + 2b + 3a + 4b$ **(b)** $7x + 3y - 2x + y$ **(c)** $9m - 4n - 3m + 6n$

(d) $3p + 5 + 2p - 8$ **(e)** $8k - k + 2h - 5h$ **(f)** $4x^2 + 3x - x^2 + 2x$

C8 Expanding brackets

KEY IDEAS

Multiply the term outside by every term inside

$a(b + c) = ab + ac$ Draw arrows from the outside term to **each** term inside. Watch the signs: a negative outside changes the sign of every term inside.

WORKED EXAMPLE

(a) $4(x + 3) = 4 \times x + 4 \times 3 =$ **$4x + 12$** (b) $5(2a - 1) =$ **$10a - 5$** (c) $-2(y + 6) =$ **$-2y - 12$**

IN THE EXAM Section C, Q19(b) (1 mark)

Practice – expand

(a) $2(x + 5)$ **(b)** $6(a + 2)$ **(c)** $3(m - 4)$ **(d)** $5(2x + 3)$

(e) $4(3y - 2)$ **(f)** $-3(k + 7)$ **(g)** $a(a + 6)$ **(h)** $2x(x - 5)$

C9 Substituting into expressions

KEY IDEAS

Replace each pronumeral with its value, in brackets

$2m^2$ means $2 \times m^2$: **square first**, then multiply by 2. It is **not** $(2m)^2$.

$3n$ means $3 \times n$. Use the order of operations: powers, then × and ÷, then + and −.

WORKED EXAMPLE

Evaluate $3a^2 + 2b$ when $a = 5$ and $b = 4$.

$= 3 \times (5)^2 + 2 \times (4) = 3 \times 25 + 8 = 75 + 8 =$ **83**

IN THE EXAM Section C, Q19(c) (1 mark)

Practice – evaluate each expression when $x = 3$, $y = 6$ and $z = -2$

(a) $4x + y$ **(b)** $5y - 2x$ **(c)** x^2 **(d)** $2y^2$ **(e)** $3x^2 - y$

(f) $xy + z$ **(g)** z^2 **(h)** $y^2 \div x$ **(i)** $4x^2 - 5z$

Measurement

C10 Area of a trapezium

KEY IDEAS

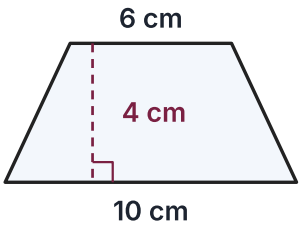
A = ½ × (a + b) × h

a and *b* are the two **parallel** sides. *h* is the **perpendicular** height – the distance straight across between the parallel sides, not the slanted side.

Add the parallel sides first (brackets!), then multiply by the height and halve. Area units are squared: cm², m².

WORKED EXAMPLE

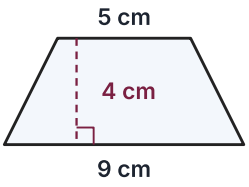
A = ½ × (6 + 10) × 4
= ½ × 16 × 4
= **32 cm²**



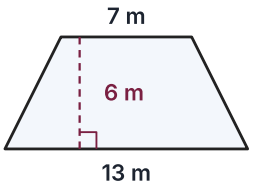
IN THE EXAM Section C, Q20(a) (2 marks)

Practice – find the area of each trapezium

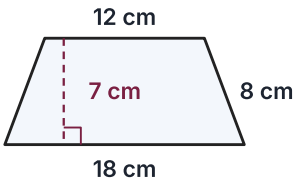
(a)



(b)



(c)



(d) A trapezium has parallel sides of 3 cm and 8 cm and a perpendicular height of 5 cm. Find its area.

(e) A garden bed is a trapezium with parallel sides 2.5 m and 4.5 m, which are 2 m apart. Find its area.

C11 Volume of a rectangular prism

KEY IDEAS

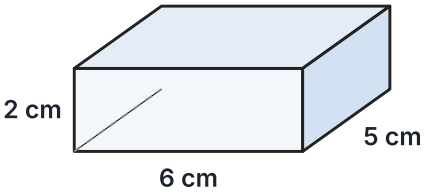
V = length × width × height

Multiply the three dimensions. Make sure they are all in the same unit first. Volume units are cubed: cm^3, m^3.

Useful fact: 1000 cm^3 = 1 litre.

WORKED EXAMPLE

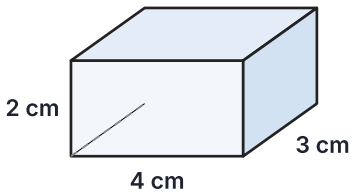
$V = 6 \times 5 \times 2$
$= \mathbf{60\ cm^3}$



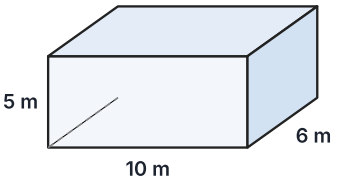
IN THE EXAM Section C, Q20(b) (2 marks)

Practice

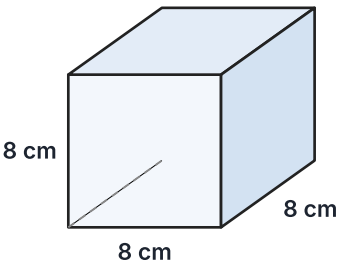
(a)



(b)



(c)



(a) Find the volume of a prism 12 cm long, 2.5 cm wide and 4 cm high.

(b) A box is 20 cm × 15 cm × 10 cm. Find its volume in cm^3 and in litres.

(c) A prism has a volume of 120 cm^3, a length of 6 cm and a width of 5 cm. Find its height.

Linear relationships, ratios and rates

C12 Tables of values for a linear rule

KEY IDEAS

Substitute every x-value – use brackets for negatives

For $y = 2x - 3$ when $x = -1$: $y = 2 \times (-1) - 3 = -2 - 3 = -5$

For a linear rule, the **y**-values go up (or down) by the **same amount** each time – the number in front of x. Use this pattern to check your table. Plotted points lie in a straight line.

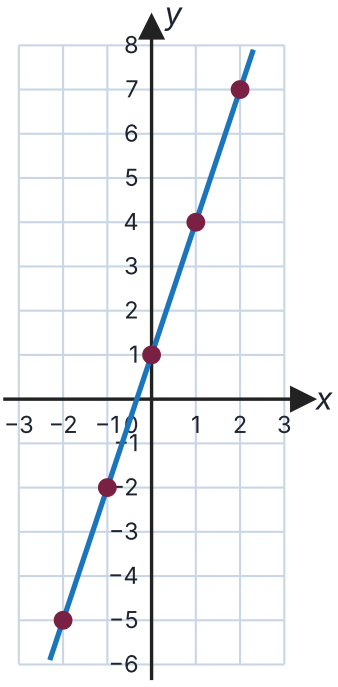
WORKED EXAMPLE

Complete the table for $y = 3x + 1$.

x	−2	−1	0	1	2
y	−5	−2	1	4	7

e.g. $x = -2$: $3 \times (-2) + 1 = -6 + 1 = -5$

Each y-value is 3 more than the one before ✓



IN THE EXAM Section C, Q21 (2 marks)

Practice – complete each table of values

(a) $y = x + 4$

x	−2	−1	0	1	2
y					

(b) $y = 2x + 1$

x	−2	−1	0	1	2
y					

(c) $y = 3x - 2$

x	−2	−1	0	1	2
y					

(d) $y = 4 - x$

x	−2	−1	0	1	2
y					

(e) $y = 2x - 5$

x	−1	0	1	2	3
y					

(f) $y = 5x$

x	−2	−1	0	1	2
y					

C13 Simplifying ratios

KEY IDEAS

Divide both parts by the highest common factor

Simplifying a ratio is like simplifying a fraction: divide every part by the **HCF**. Keep the order the same as the question. Convert to the **same units** first and leave units out of the answer.

WORKED EXAMPLE

(a) 12 : 18 → HCF is 6 → 12 ÷ 6 : 18 ÷ 6 = **2 : 3**

(b) 40 cm : 2 m → 40 cm : 200 cm → ÷ 40 → **1 : 5**

IN THE EXAM Section C, Q22(a) (1 mark)

Practice – write in simplest form

(a) 6 : 9 **(b)** 10 : 25 **(c)** 16 : 20 **(d)** 21 : 35

(e) 36 : 48 **(f)** 45 : 30 **(g)** 50 cm : 1 m

(h) A class has 28 students and 12 are boys. Write the ratio boys : girls in simplest form.

C14 Average speed

KEY IDEAS

Speed = distance ÷ time

Average speed = $\frac{\text{distance}}{\text{time}}$. For km/h, distance must be in km and time in **hours** (30 minutes = 0.5 h, 15 minutes = 0.25 h).

Rearranged: distance = speed × time, time = distance ÷ speed.

WORKED EXAMPLE

(a) A car travels 150 km in 2 hours. Speed = 150 ÷ 2 = **75 km/h**

(b) A train travels 45 km in 30 minutes. 30 min = 0.5 h → 45 ÷ 0.5 = **90 km/h**

IN THE EXAM Section C, Q22(b) (1 mark)

Practice

(a) A bus travels 240 km in 4 hours. Find its average speed.

(b) A plane flies 360 km in 3 hours. Find its average speed.

(c) A cyclist rides 18 km in 2 hours. Find her average speed.

(d) A car travels 100 km in 1.25 hours. Find its average speed.

(e) A ferry travels 40 km in 30 minutes. Find its average speed in km/h.

(f) A train travels at 90 km/h for 3 hours. How far does it travel?

(g) How long does it take to travel 200 km at 80 km/h?

The night before and on the day

Formulas and facts to know by heart

- Inverse operations: + ↔ −, × ↔ ÷; undo in reverse order
- Check: substitute back in; LHS = RHS
- Amount from a sector graph = % × total
- Mean = sum ÷ number of values
- Median: order first; even number → average the middle two
- Range = highest − lowest
- $c^2 = a^2 + b^2$ (c is the hypotenuse)
- $\sqrt{a \times b} = \sqrt{a} \times \sqrt{b}$
- Trapezium: A = ½ × (a + b) × h
- Rectangular prism: V = l × w × h
- Speed = distance ÷ time

Common mistakes to avoid

- Undoing × or ÷ before + or − in two-step equations
- Losing the negative sign: 4 − 10 = −6, not 6
- Adding the *values* row instead of the *frequency* row
- Giving the frequency as the mode instead of the value
- Counting 40 in “more than 40”
- Finding the median without ordering the data first
- Writing the range as “3 to 12” instead of 9
- $2^4 = 8$ (it is 16), and $2m^2$ treated as $(2m)^2$
- Forgetting the square root in Pythagoras
- Using the slanted side as the height of a trapezium
- Time in minutes instead of hours for km/h

Your 75-minute plan

0–25 min	25–50 min	50–75 min
Section A Equations	Section B Investigating Data	Section C Mixed Topics

Aim to finish each section a couple of minutes early so you have about 5 minutes at the end to go back to any skipped questions and check your answers.

Answers

Section A · Equations

A1 (a) 7 (b) 16 (c) 15 (d) 21 (e) 9 (f) 8 (g) 27 (h) 30

A2 (a) 5 (b) 7 (c) 6 (d) 8 (e) 12 (f) 20 (g) 9 (h) 5

A3 (a) −5 (b) −7 (c) −5 (d) −4 (e) −3 (f) −4 (g) −10 (h) −3

A4 (a) LHS = 17 = RHS → yes (b) LHS = 12 ≠ 14 → no (c) LHS = 7 = RHS → yes (d) LHS = 4 = RHS → yes (e) LHS = 30 ≠ 32 → no (f) LHS = 1 = RHS → yes

A5 (a) n + 15 = 32, n = 17 (b) 5n + 4 = 39, n = 7 (c) 6n − 10 = 44, n = 9 (d) n ÷ 2 + 9 = 15, n = 12 (e) 3n − 5 = −11, n = −2

A6 (a) 3n + 8 = 23, n = 5; one notebook costs \$5 (b) 2k + 4 = 26, k = 11; the trip was 11 km (c) 10v + 15 = 85, v = 7; 7 visits (d) c ÷ 6 + 2 = 9, c = 42; the pizza order cost \$42 (e) 2w + 18 = 30, w = 6; the width is 6 cm

Section B · Investigating Data

B1 (a) Chocolate (b) Mint (c) 35 (d) 11 − 5 = 6 (e) 7/35 = 1/5

B2 (a) 1 pm (b) 50 (c) 80 − 20 = 60 (d) 95 − 45 = 50 (e) between 11 am and 12 pm (a rise of 30)

B3 (a) Maths (b) 0.25 × 80 = 20 (c) 0.15 × 80 = 12 (d) 24 − 8 = 16 (e) 40% of 80 = 32 (f) 80% of 80 = 64

B4 (a) 13 (b) 45 (c) 7th value = 29 (d) 45 − 14 = 31 (e) 6 (f) 9

B5 (a) 5 (b) 3 visits (c) 4 + 2 + 1 = 7 (d) 2 + 1 = 3 (e) 2 + 4 = 6

B6 (a) 4 (b) 7 (c) 3 (d) 5 (e) 2 (f) 3

B7 (a) 24 (b) 2 goals (c) 3 (d) 4 + 2 + 1 = 7 (e) 8 + 4 + 2 + 1 = 15 (f) 3 + 6 = 9

B8 (a) 35 ÷ 5 = 7 (b) 75 ÷ 5 = 15 (c) 30 ÷ 6 = 5 (d) 138 ÷ 6 = 23 (e) 31 ÷ 5 = 6.2 (f) 119 ÷ 8 = 14.875 ≈ 14.9

B9 (a) 5.5 (b) 10.5 (c) 18 (d) 5.5 (e) 29 (f) 13 (odd number of values)

B10 (a) mode 4, range 5 (b) mode 22, range 19 (c) mode 3.1, range 0.9 (d) modes 10 and 12, range 8 (e) no mode, range 22

B11 1. (a) \$30 (b) \$72 ÷ 6 = \$12 (c) \$42 ÷ 5 = \$8.40 (d) the outlier pulls the mean up by \$3.60, so \$12 is higher than five of the six prices. 2. (a) 3 (b) 112 ÷ 7 = 16 (c) 109 ÷ 6 = 18.2 (d) median = 18 both times – the outlier changed the mean but not the median

Section C · Mixed Topics

C1 (a) 10 cm (b) 15 m (c) $\sqrt{65}$ ≈ 8.1 cm (d) $\sqrt{325}$ ≈ 18.0 cm (e) $\sqrt{8425}$ ≈ 91.8 cm

C2 (a) 169 = 25 + 144 → yes (b) 121 ≠ 36 + 64 = 100 → no (c) 1369 = 144 + 1225 → yes (d) 676 = 100 + 576 → yes (e) 49 ≠ 16 + 36 = 52 → no (f) 1681 = 81 + 1600 → yes

C3 (a) 1.5 (b) 0.36 (c) 1.6 (d) 0.056 (e) 8 (f) 90 (g) 120 (h) 25

C4 (a) 32 (b) 27 (c) 10 000 (d) 125 (e) 6 (f) 11 (g) 12 (h) 16 − 5 = 11 (i) 9 (j) 10 × 8 = 80

C5 (a) $\sqrt{5}$, $\sqrt{30}$ (b) $\sqrt{18}$, $\sqrt{2}$ (c) $\sqrt{45}$, $\sqrt{15}$ (d) $\sqrt{72}$, $\sqrt{3}$

C6 (a) 2$\sqrt{2}$ (b) 3$\sqrt{2}$ (c) 2$\sqrt{5}$ (d) 3$\sqrt{3}$ (e) 3$\sqrt{5}$ (f) 4$\sqrt{3}$ (g) 5$\sqrt{3}$ (h) 7$\sqrt{2}$ (i) 10$\sqrt{2}$

C7 (a) 8a + 6b (b) 5x + 4y (c) 6m + 2n (d) 5p − 3 (e) 7k − 3h (f) 3x^2 + 5x

C8 (a) 2x + 10 (b) 6a + 12 (c) 3m − 12 (d) 10x + 15 (e) 12y − 8 (f) −3k − 21 (g) a^2 + 6a (h) 2x^2 − 10x

C9 (a) 18 (b) 24 (c) 9 (d) 72 (e) 21 (f) 16 (g) 4 (h) 12 (i) 46

C10 (a) ½ × 14 × 4 = 28 cm² (b) ½ × 20 × 6 = 60 m² (c) ½ × 30 × 7 = 105 cm² (the 8 cm slanted side is not needed) (d) 27.5 cm² (e) 7 m²

C11 (a) 24 cm³ (b) 300 m³ (c) 512 cm³ (d) 120 cm³ (e) 3000 cm³ = 3 L (f) 120 ÷ 30 = 4 cm

C12 (a) 2, 3, 4, 5, 6 (b) −3, −1, 1, 3, 5 (c) −8, −5, −2, 1, 4 (d) 6, 5, 4, 3, 2 (e) −7, −5, −3, −1, 1 (f) −10, −5, 0, 5, 10

C13 (a) 2 : 3 (b) 2 : 5 (c) 4 : 5 (d) 3 : 5 (e) 3 : 4 (f) 3 : 2 (g) 50 : 100 = 1 : 2 (h) 12 : 16 = 3 : 4

C14 (a) 60 km/h (b) 120 km/h (c) 9 km/h (d) 80 km/h (e) 80 km/h (f) 270 km (g) 2.5 hours (2 h 30 min)